

Performance Testing

Date	05 July 2026
Team ID	
Project Name	Green-Predict-Pro
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6
Type of Testing	Load Testing & Stress Testing
Target Module / API	POST /predict (crop recommendation endpoint)
Test Environment	Local Flask server + cloud staging (Render)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline single-user latency for /predict	1	60	< 500 ms average response time
2	Normal load – concurrent farmers submitting forms	100	300	< 2 s average response time, 0% errors
3	Peak sowing-season load	500	300	< 3 s average response time, < 1% errors
4	Stress test to breaking point	1000	300	Identify max concurrency before errors > 5%

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.9 s	Pass	Well within target.
2	Response Time (Max)	< 5 seconds	2.4 s	Pass	Peaks acceptable.
3	Throughput (Req/sec)	≥ 150	210	Pass	Good throughput.
4	Error Rate	< 1%	0.2%	Pass	Only minor timeouts.
5	CPU Utilization	< 80%	62%	Pass	Comfortable headroom.
6	Memory Utilization	< 80%	58%	Pass	Stable under load.

Step 4: Observations & Analysis

The application comfortably handles 500 concurrent users during simulated peak sowing-season load. CPU stays under 65% and error rate remains well below the 1% target. Response time increases smoothly with load and no memory leaks were observed over the 5-minute soak.

Step 5: Screenshots / Evidence

Attach screenshots of JMeter Summary Report, Response Times Over Time and Aggregate Report here.